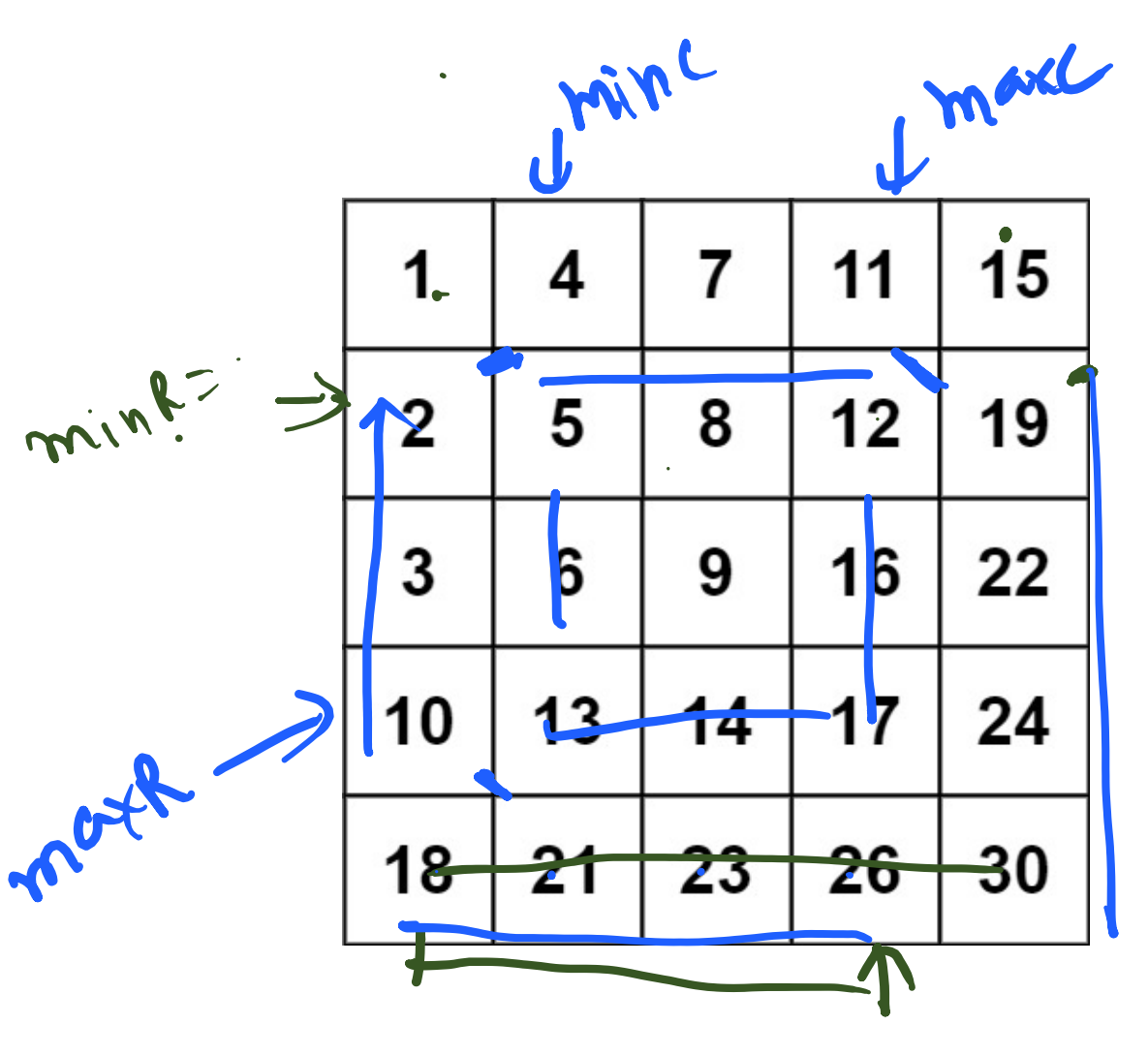
Lec-13

11/10/26 4:46 PM



minc
maxc
minR=
maxR

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

1 4 7 11 15 19 22 24 30 26 23 21 18

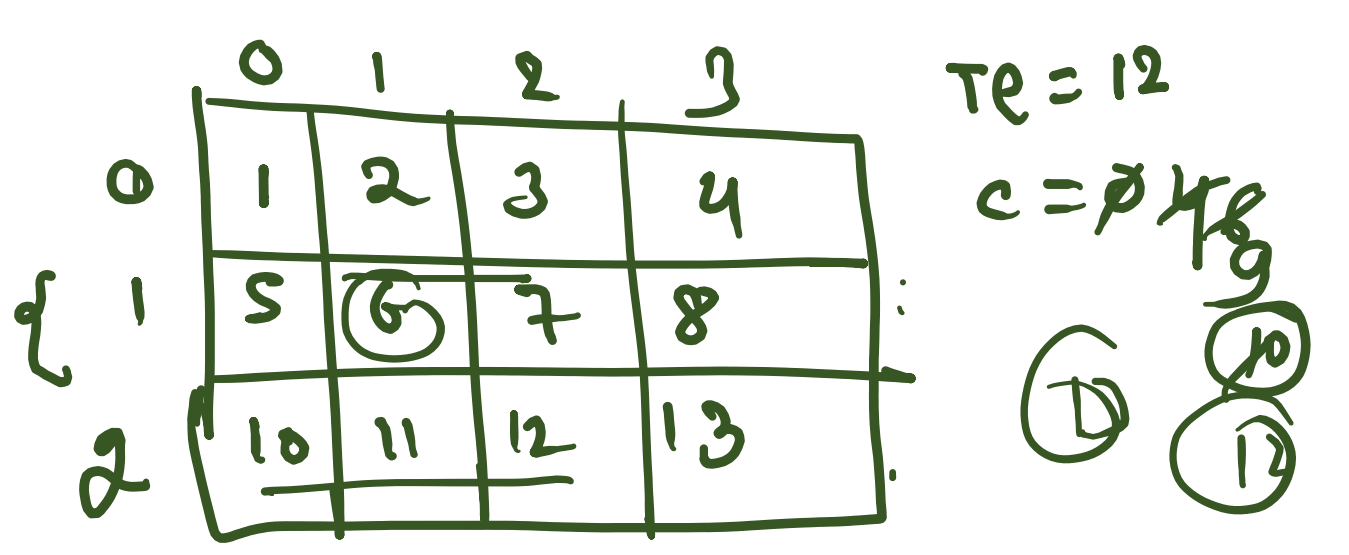
3 2 5 8 12 16 17 14 13 6 9

```
for(i=minc i<=maxC i++){
  c++
  syso(arr[minR][i])
}
minR++
for(i=minR i<=maxR i++){
  c++
  syso(arr[i][maxc])
}
maxc--
for(i=maxc i>=minc i--){
  c++
  syso(arr[maxR][i])
}
maR--
for(i=maxR i>=minR i--){
  c++
  syso(arr[i][minc])
}
minc++
```

```java
public static void Print(int[][] arr) {
	int minr = 0, minc = 0, maxr = arr.length - 1, maxc = arr[0].length - 1;
	int te = arr.length * arr[0].length;
	int c = 0;
	while (c < te) {
		for (int i = minc; i <= maxc; i++) {
			c++;
			System.out.print(arr[minr][i] + " ");
		}
		minr++;
		for (int i = minr; i <= maxr; i++) {
			c++;
			System.out.print(arr[i][maxc] + " ");
		}
		maxc--;
		for (int i = maxc; i >= minc; i--) {
			c++;
			System.out.print(arr[maxr][i] + " ");
		}
		maxr--;
		for (int i = maxr; i >= minr; i--) {
			c++;
			System.out.print(arr[i][minc] + " ");
		}
		minc++;
	}
}
```

10<12

c=te



Te=12

c=0 4 8 9

min=0 1 2 maxR=1 0

minc=0 1 maxc=3 2 1

1 2 3 4 8 3 12 11 10 5 6 7 6

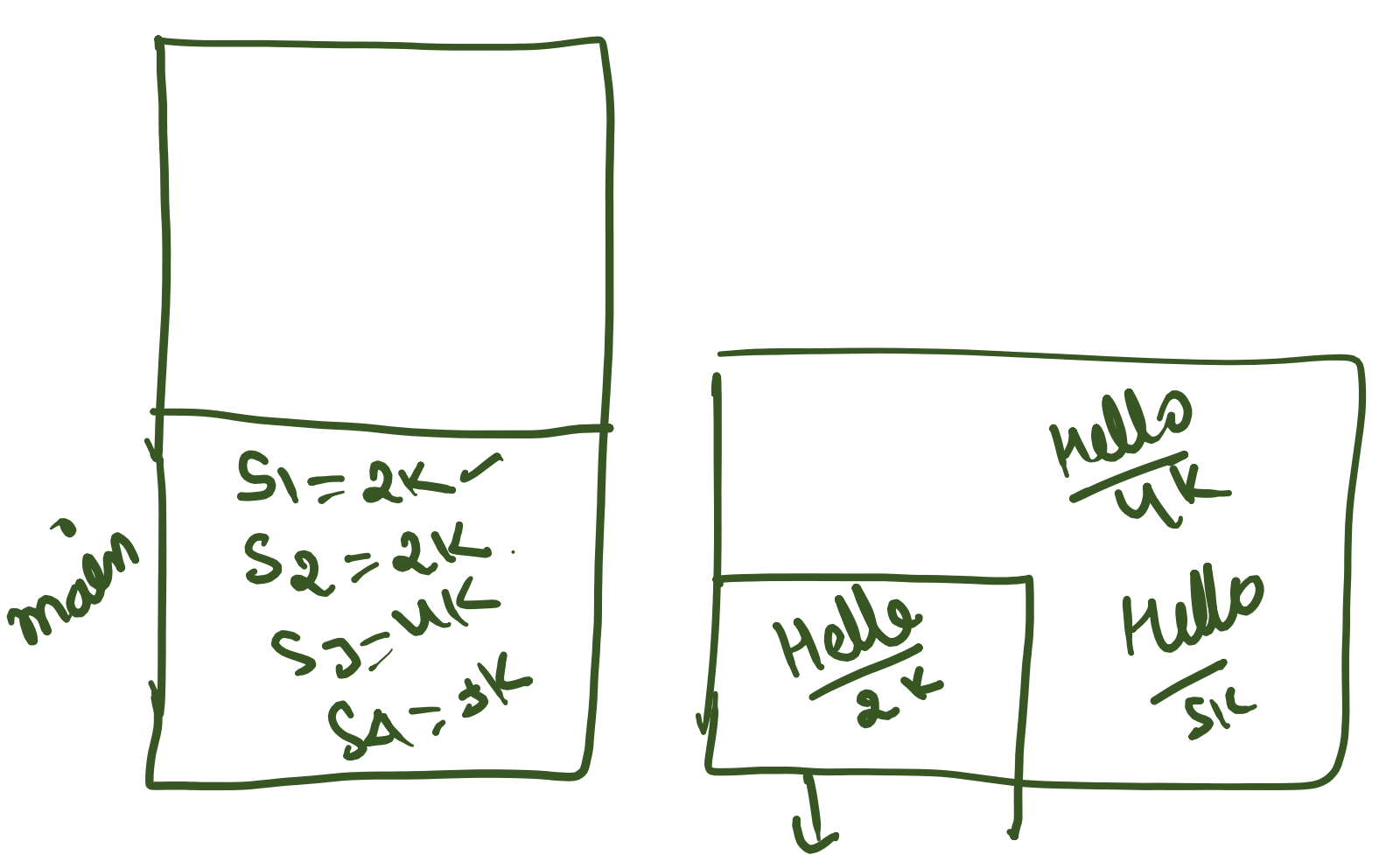
String

→ class → N.P → heap

→ immutable

Memory Design

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	String s1="Hello";
	String s2="Hello";
	String s3=new String("Hello");
	String s4=new String("Hello");
}
```

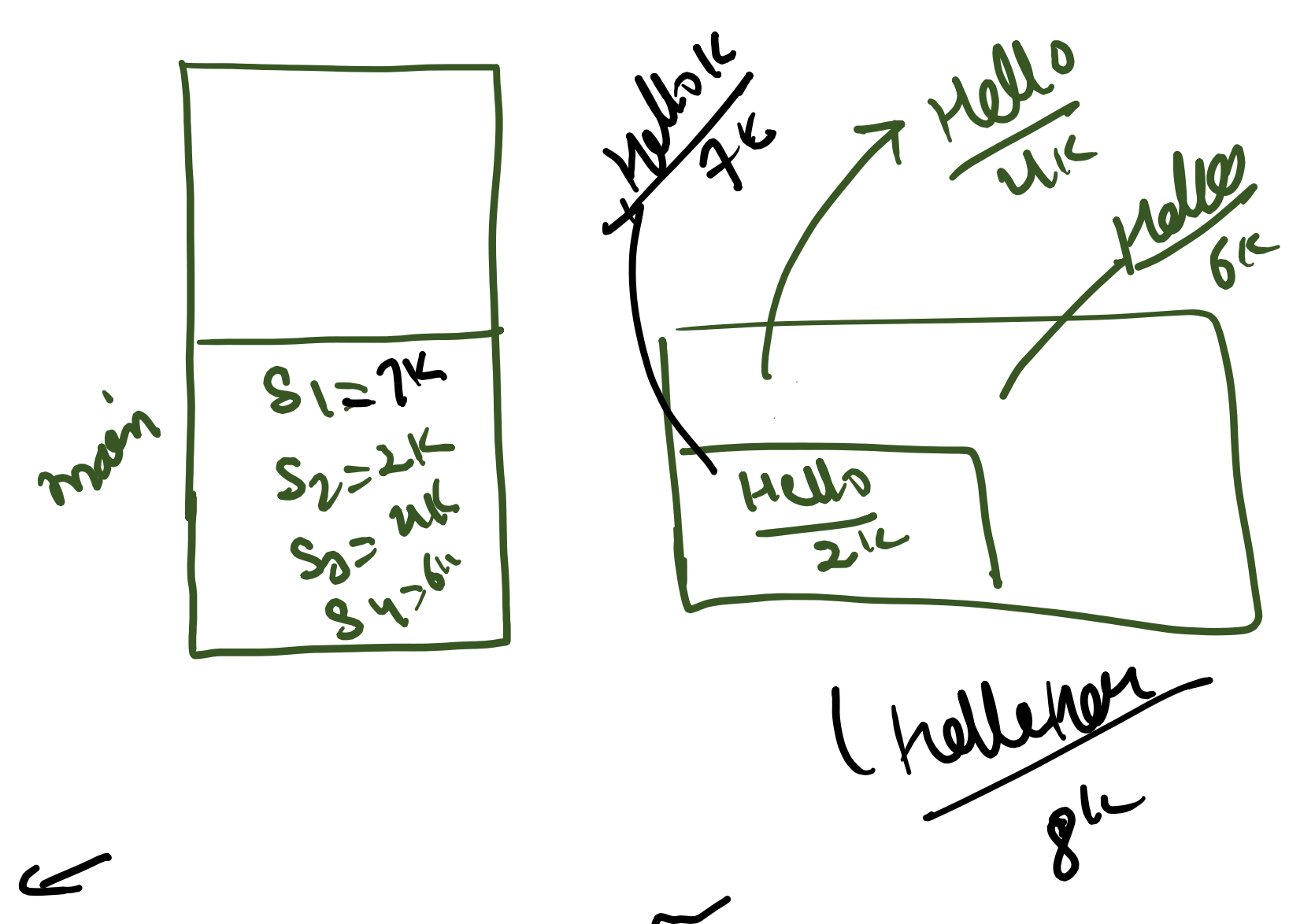


String pool

Intern pool

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	String s1 = "Hello";
	String s2 = "Hello";
	String s3 = new String("Hello");
	String s4 = new String("Hello");
	s1 = s1 + "k";
}
```

s2 = s2 + s4



nitin

naman

naan

maam

nitin

i i j j

raj

aajesh

0 X

kunal

Komal

6 s 1

Ankita

Ankit

Raj

Rajesh

aaj

Raj

aj

0 1 2 3 4

hello

i, j

0,1 h

0,2 he

0,3 hel

0,4 hell

0,5 hello

1,2 e

1,3 el

1,4 ell

1,5 ello

2,3 l

2,4 ll

2,5 llo

3,4 l

3,5 lo

4,5 o

the sky is blue → split

| the | sky | is | blue |

blue is sky the

ans=3 6

ans=3 4 5 6

k pm aei mnoap kl s aieou a lmpaq

ans=0

c=0

```
for(i=0 i<s.len() i++){
  char ch= s.charAt(i)
  if(Isvowel(ch)){
    c++
    ans= max(ans,c)
  }
  else{
    c=0
  }
}
```

hello

0 1 2 3 4

i j

[0,1] h

[1,2] e

[2,3] l

[3,4] l

[4,5] o

j-i=1

i j

0,2 he

1,3 el

2,4 ll

3,5 lo

j-i=2

i j

0,3 hel

1,4 ell

2,5 llo

j-i=3

0,4 hell

1,5 ello

j-i=4

0,5 hello

j-i=5

j-i=len

i=j-len

```
for(len=1 len<=s.len() len++){
  for(j=len j<=s.len() j++){
    int i=j-len
  }
}
```